IBVAP — SIH Presentation Structure & Slide Deck Guide

15. Complete 15-Slide Presentation Deck

Slide 1: Title & Team Overview

- **Title:** IBVAP — Intelligent Border Video Analytics Platform
- **Subtitle:** AI-Powered Border Surveillance, Intrusion Detection & Cryptographic Security Analytics
- **Category:** SIH 2026 — Blockchain & Cybersecurity
- **Visual:** IBVAP Logo, Command Center Header Graphic, Team Name & ID.
- **Spoken Script:** *"Respected Judges, good morning. We are excited to present IBVAP—an Intelligent Border Video Analytics Platform designed to automate perimeter defense using AI and secure security logs using cryptographic blockchain integrity."*

Slide 2: Problem Statement & National Security Challenge

- **Bullet Points:**
- Manual surveillance of vast national borders is inefficient and prone to operator fatigue.
- Traditional CCTV requires continuous human monitor gazing, leading to delayed response.
- Standard digital audit logs can be corrupted or deleted by unauthorized internal actors.
- **Visual:** Diagram comparing manual CCTV monitor room vs automated AI alert workflow.
- **Spoken Script:** *"Border security forces monitor thousands of camera feeds 24/7. Human fatigue creates blind spots. Furthermore, if a breach occurs, traditional logs stored in plain databases can be altered or erased. IBVAP solves both challenges."*

Slide 3: Proposed IBVAP Solution Overview

- **Bullet Points:**
- **Automated Detection & Tracking:** YOLO11 neural detection + ByteTrack target persistence.
- **Geofenced Perimeter Defense:** Real-time polygon restricted zone intersection detection.
- **Instant Forensic Evidence:** Auto-capture of breach snapshots linked to event metadata.
- **Tamper-Evident Security:** SHA-256 linear hash chaining and block-linked audit ledger.
- **Visual:** High-level 4-pillar solution diagram.
- **Spoken Script:** *"IBVAP combines real-time AI computer vision with cryptographic security. It instantly detects intruders, tracks them across occlusions, captures high-resolution evidence, and writes every event into a tamper-evident audit ledger."*

Slide 4: System Architecture & Data Pipeline

- **Bullet Points:**
- End-to-end pipeline: Input Feed $\rightarrow$ YOLO11 $\rightarrow$ ByteTrack $\rightarrow$ Zone Polygon $\rightarrow$ Evidence $\rightarrow$ MongoDB $\rightarrow$ SHA-256 Audit Chain $\rightarrow$ Streamlit Dashboard.
- Multi-tier database resilience (MongoDB Atlas $\rightarrow$ CSV $\rightarrow$ Fallback Seed Data).
- **Visual:** Flowchart diagram of complete system pipeline (from `02_ARCHITECTURE.md`).
- **Spoken Script:** *"Our architecture handles everything from low-level frame processing to cloud database sync. Even if network connectivity fails, our multi-tier fallback ensures 100% operational uptime."*

Slide 5: AI & Computer Vision Engine

- **Bullet Points:**
- **Object Detection:** Ultralytics YOLO11 (`yolo11s.pt` / `yolo11n.pt`) tuned for human detection (`class 0`).
- **Multi-Object Tracking:** ByteTrack tracker using high/low confidence score fusion (`track_high_thresh: 0.25`, `track_low_thresh: 0.10`).
- **Occlusion Handling:** Kalman filter prediction and 60-frame track buffer.
- **Visual:** Annotated surveillance frame displaying bounding boxes and Track IDs (`T:2`, `T:7`).
- **Spoken Script:** *"We chose YOLO11 for speed and accuracy, paired with ByteTrack. ByteTrack is special because it doesn't discard low-confidence detections, allowing us to keep tracking intruders even when they pass behind obstacles."*

Slide 6: Geofenced Intrusion Detection Logic

- **Bullet Points:**
- Dynamic 2D spatial polygon coordinates representing restricted border corridors.
- Centroid intersection algorithm determines `ENTER`, `EXIT`, and `INITIAL_INSIDE` states.
- Automated deduplication prevents repetitive alerts for the same tracked target.
- **Visual:** Restricted zone polygon diagram showing target entering red perimeter.
- **Spoken Script:** *"When a person's tracked bounding box crosses our defined polygon boundary, the system instantly triggers an 'ENTER' breach event, assigns a HIGH risk rating, and freezes an evidence frame."*

Slide 7: Digital Evidence & Visual Snapshots

- **Bullet Points:**
- High-resolution evidence image capture at the exact frame of breach.
- Automatic file naming (`EVT-XXX_track_Y_frame_Z.jpg`) mapping event to track ID.
- Interactive evidence gallery with zoom viewer and instant download capability.
- **Visual:** Screenshot of IBVAP Digital Evidence gallery page (`EVT-001`, `EVT-002`, `EVT-003`).
- **Spoken Script:** *"Here you see our digital evidence manager. Every breach auto-generates a forensic snapshot that security commanders can inspect in high-resolution or download for legal handoff."*

Slide 8: Database Architecture & Multi-Tier Resilience

- **Bullet Points:**
- MongoDB Atlas NoSQL collection (`IBVAP.intrusion_events`).
- Standardized JSON schema for event payloads.
- 3-Tier fallback strategy: MongoDB Cloud $\rightarrow$ Local CSV (`intrusion_log.csv`) $\rightarrow$ Seed Data.
- **Visual:** Diagram showing 3-tier fallback loader flow.
- **Spoken Script:** *"We integrated MongoDB Atlas for cloud flexibility. If the cloud database goes offline due to border network disruptions, IBVAP seamlessly switches to local CSV logging without losing a single frame."*

Slide 9: SHA-256 Cryptographic Audit Chain

- **Bullet Points:**
- Each event string is concatenated with the previous record's SHA-256 digest.
- Root hash stored separately in `audit_root_hash.txt`.
- One-click live verification detects modified, deleted, or inserted logs instantly.
- **Visual:** Hash chain diagram ($Record_1 \rightarrow Hash_1 \rightarrow Record_2 + Hash_1 \rightarrow Hash_2$).
- **Spoken Script:** *"To ensure cybersecurity integrity, we hash-chain every record using SHA-256. If a malicious insider tries to change a record's Track ID or timestamp, the hash chain breaks instantly, alerting commanders to tampering."*

Slide 10: Local Blockchain Audit Ledger

- **Bullet Points:**
- Single-node tamper-evident JSON ledger (`blockchain_ledger.json`).
- Block 0 initialized as Genesis Block (`previous_hash` = 64 zeros).
- Event payloads hashed into sequential block headers.
- Architected for future enterprise expansion to Hyperledger Fabric.
- **Visual:** JSON block structure snippet showing Block 0 Genesis and Block 1 Event payload.
- **Spoken Script:** *"We have also built a local block-linked ledger. Each security event forms a block linked to the previous block's digest, providing a working proof-of-concept for blockchain-backed auditing."*

Slide 11: Web Command Center Dashboard

- **Bullet Points:**
- Built using Streamlit, Plotly, and custom Dark Navy UI components.
- 6 Operational views: Command Center, Surveillance, Security Events, Digital Evidence, Audit Log, System Page.
- Real-time KPI metrics, Plotly threat charts, and live system health indicators.
- **Visual:** Side-by-side screenshot of Command Center and Surveillance views.

- **Spoken Script:** *"Our Command Center gives operators full situational awareness. It includes live threat feeds, interactive timelines, evidence zoom, and real-time audit chain verification."*

Slide 12: Deployment & Cloud Infrastructure

- **Bullet Points:**
- Deployed on Streamlit Community Cloud with GitHub CI/CD integration.
- Environment secrets (`MONGODB_URI`, `SUPABASE_KEY`) managed via Streamlit Secret Vault.
- Accessible via standard web browsers across mobile and desktop devices.
- **Visual:** Cloud deployment architecture topology map.
- **Spoken Script:** *"IBVAP is fully deployed and accessible via public web URL. All sensitive database connection strings are protected inside encrypted cloud secret vaults."*

Slide 13: Innovation & Competitive Advantage

- **Bullet Points:**
- Combines computer vision with cryptographic cybersecurity auditability.
- Score fusion tracking minimizes false-positive alerts.
- Zero-downtime multi-tier data fallback architecture.
- Fully working end-to-end prototype tested on surveillance feeds.
- **Visual:** Comparison table of IBVAP vs Traditional CCTV Systems.
- **Spoken Script:** *"What sets IBVAP apart is the fusion of AI vision with cryptographic security. We don't just detect intrusions—we protect the integrity of the evidence."*

Slide 14: Limitations & Future Scope

- **Bullet Points:**
- **Limitations:** Single-camera feed ingestion in current prototype; GPU/TensorRT acceleration unmeasured.
- **Future Roadmap:** Multi-camera RTSP ingestion, thermal IR sensor fusion, Hyperledger Fabric enterprise deployment.
- **Visual:** Milestone roadmap graphic (Phase 1 Current $\rightarrow$ Phase 2 Multi-Camera $\rightarrow$ Phase 3 Enterprise Blockchain).
- **Spoken Script:** *"We are honest about our current prototype scope. Today we demonstrate a robust single-stream platform, and our roadmap includes scaling to multi-camera thermal streams and multi-node permissioned blockchains."*

Slide 15: Conclusion & Q&A

- **Bullet Points:**
- IBVAP provides intelligent, automated, and tamper-proof border surveillance.
- Fully deployed, tested, and verified system.

- Thank you! We welcome your questions.
- **Visual:** Team contact info, GitHub repo link, QR code to live web app.
- **Spoken Script:** *"Thank you, Honorable Judges. IBVAP is ready to strengthen our nation's border security. We are now open for your technical questions."*